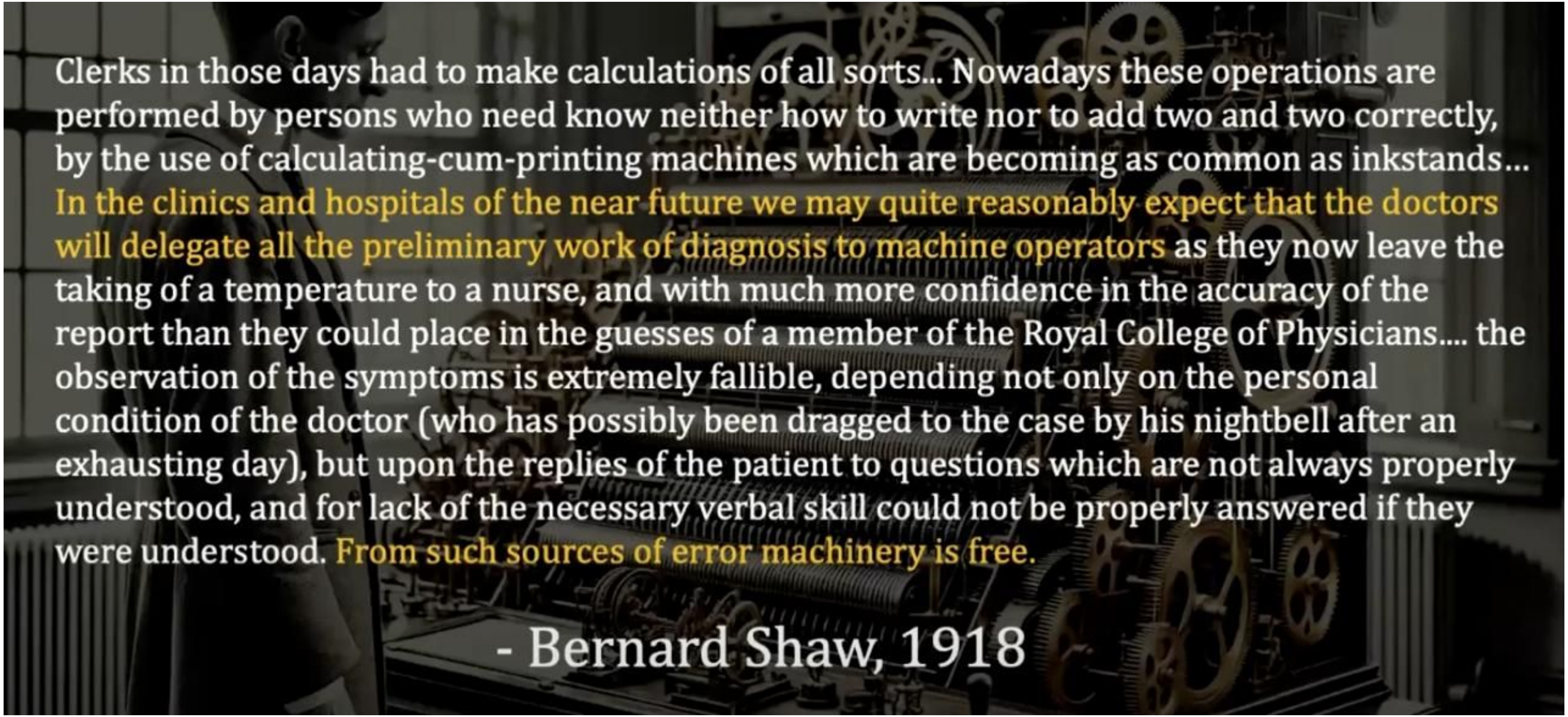
Applications of AI Agents in Medicine and Healthcare

Transforming Medical Education, Clinical Practice, and Patient Care

25th February 2026, KMC Manipal

Ashish Makani
KCDH-A, Ashoka University

AI in medicine has very old roots !




2015
SELF-DRIVING CARS IN 2 YEARS
CausalPython.io
imgflip.com
2016
RADIOLOGISTS OBSOLOLETE IN 5 YEARS
2025
RADIOLOGISTS IN REGULAR CARS DRIVING TO WORK

Introduction to AI in Healthcare

Artificial Intelligence is transforming healthcare across multiple domains:

- **Clinical Decision Support:** AI systems that analyze patient data to assist in diagnosis and treatment planning
- **Medical Education:** Virtual patients and simulation tools for training healthcare professionals
- **Administrative Efficiency:** Automation of routine tasks to reduce clinician burden
- **Patient Engagement:** Tools that help patients understand and participate in their care

By 2025, over 75% of healthcare organizations have implemented or are planning to implement AI solutions to address clinical and operational challenges.

Source: Healthcare Information and Management Systems Society (HIMSS) 2025 Survey

AI in Medical Education: Overview

AI is revolutionizing how future healthcare professionals learn:

Virtual Patient Simulations

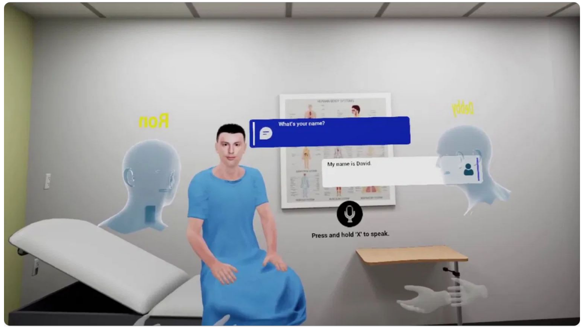
AI-powered virtual patients that respond realistically to student interactions

Adaptive Learning Systems

Personalized education pathways based on individual student performance

Performance Analytics

Real-time feedback on clinical reasoning and decision-making



"AI in medical education is not about replacing human instructors but augmenting their capabilities to provide more personalized, accessible, and effective learning experiences."

— Journal of Medical Education and Technology, 2025

AI for Learning History Taking

Virtual patients powered by AI provide medical students with realistic practice for history taking skills:

Realistic Conversations

Natural language processing enables human-like dialogue with virtual patients

Diverse Case Library

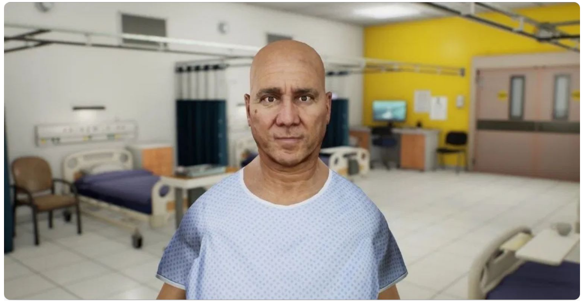
Exposure to wide range of patient presentations and medical histories

Immediate Feedback

AI analyzes questioning patterns and provides guidance on missed areas

Safe Learning Environment

Students can practice repeatedly without patient fatigue or time constraints



DDx demo


AI in Medical Education Symposium
LIGHTNING
DEMOS: AI TOOLS
FOR MEDICAL
EDUCATION

Improving Empathy & Bedside Manner

AI systems are helping medical students develop crucial interpersonal skills:

Real-time Feedback

AI analyzes tone, word choice, and body language during simulated patient interactions

Emotional Intelligence Training

Virtual patients display realistic emotional responses to different communication approaches

Personalized Coaching

AI identifies individual communication patterns and suggests specific improvements

"Students who trained with AI-enhanced empathy simulations showed a 42% improvement in patient satisfaction scores during

Enhancing Diagnostic Skills

AI tools are transforming how medical students develop diagnostic reasoning:

Case-Based Learning

AI systems present diverse clinical scenarios with varying complexity and presentations

Interactive Feedback

Real-time guidance on diagnostic reasoning process and decision points

Pattern Recognition

Training on thousands of cases to improve identification of subtle clinical patterns

Studies show students using AI-enhanced diagnostic training demonstrate 40% improvement in accuracy compared to traditional methods

AI in Radiology: Overview

Radiology faces significant challenges that AI is helping to address:

122,000

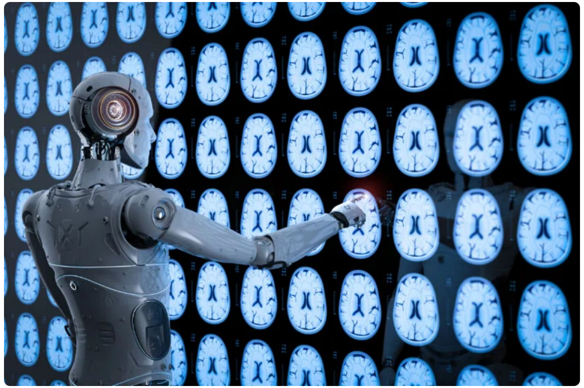
Projected physician shortage by 2032, with radiology among the most affected specialties

30%

Annual increase in imaging studies in some healthcare systems

84%

Of radiologists report symptoms of burnout, partly due to increasing workloads



Sources: AAMC Report 2022, Radiology Business Journal 2024, American College of Radiology 2025

AI applications in radiology are expanding rapidly, from

AI as a Copilot for Radiologists

AI systems are increasingly serving as intelligent assistants to radiologists:

Enhanced Detection

AI algorithms can flag potential abnormalities for radiologist review, reducing the risk of missed findings

Prioritization

Intelligent worklist sorting ensures critical cases are reviewed first

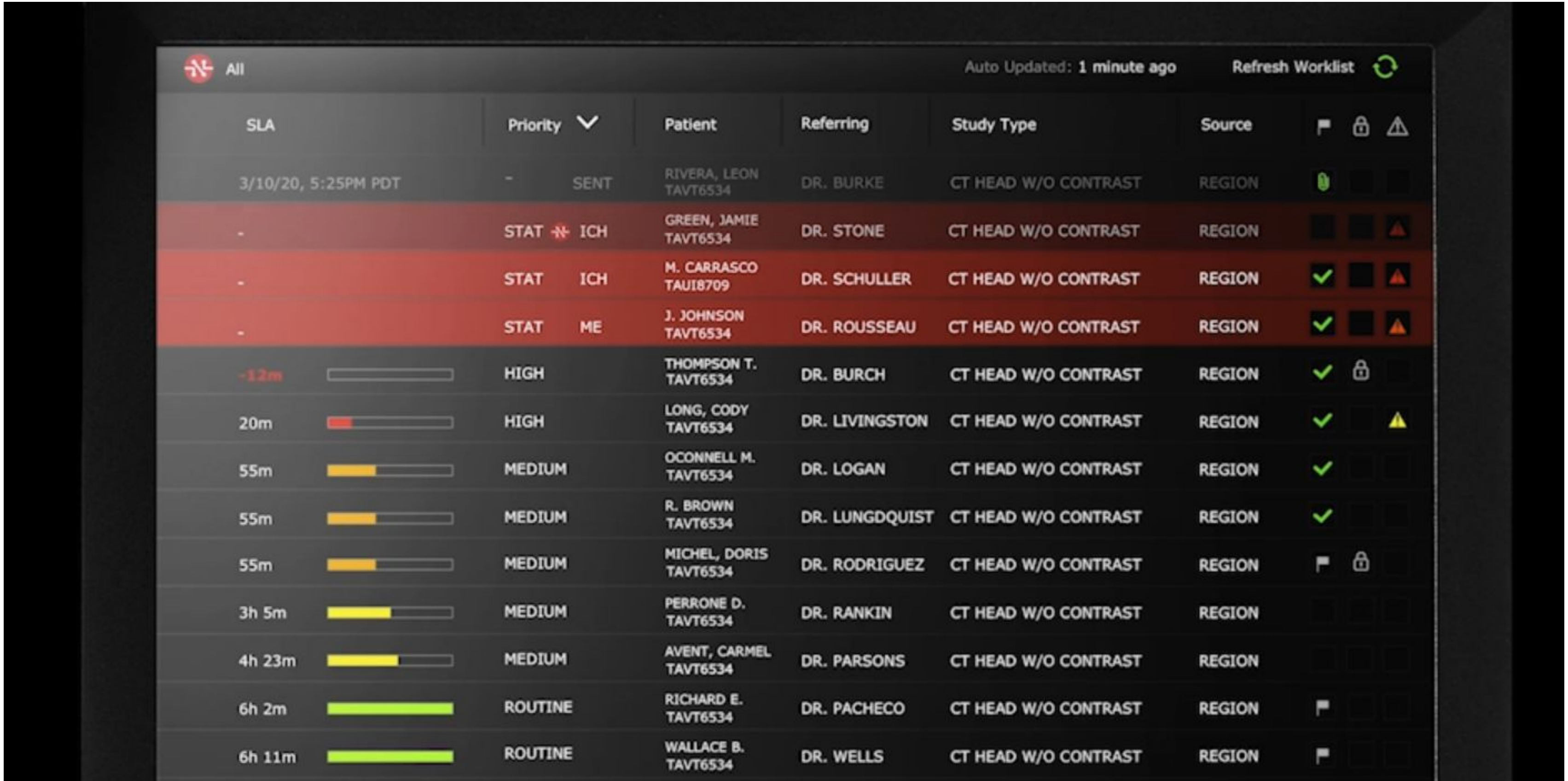
Automated Reporting

AI-assisted report generation with structured templates and measurements

Efficiency Gains

Studies show 22-29% reduction in reading time when using AI assistance

Worklist Prioritization



AI for Patient-Friendly Radiology Reports

AI is helping bridge the gap between complex medical terminology and patient understanding:

Original Radiology Text:

"Bilateral scattered fibroglandular densities with no suspicious masses, architectural distortion, or suspicious calcifications."

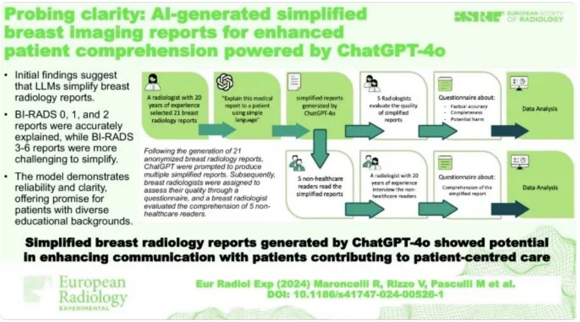
AI-Simplified Version:

"Your breast tissue appears normal. We did not find any concerning lumps or calcium deposits that might suggest cancer."

Google's RadExplain and similar tools use natural language processing to:

- Translate medical jargon into plain language
- Provide context for medical findings
- Explain next steps in patient-friendly terms

Studies show patients using AI-translated reports have 43% better understanding of their condition and 37% reduced anxiety.



Rad Demo 1

Rad Demo 2

AI for Surgical Video Analysis

AI systems are revolutionizing how surgical videos are analyzed and utilized:

Automated Phase Recognition

AI identifies critical stages in surgical procedures, enabling precise analysis and comparison

Quality Assessment

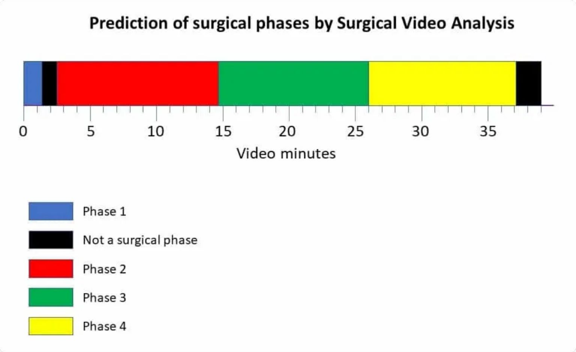
Systems like MedGemini evaluate technical aspects such as Critical View of Safety (CVS) in laparoscopic procedures

Educational Tool

AI-annotated videos serve as powerful teaching resources for surgical trainees

Continuous Improvement

Analysis of surgical techniques across multiple procedures enables evidence-based refinement



Laparoscopic Cholecystectomy CVS demo

Improving Surgical Quality with AI

AI systems like MedGemini analyze surgical videos to assess quality metrics:

Critical View of Safety (CVS) Assessment

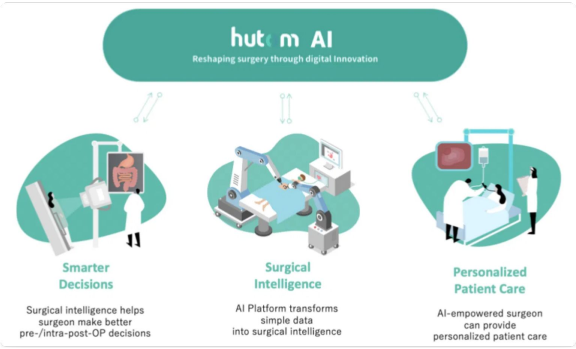
AI evaluates whether key safety criteria are met during laparoscopic cholecystectomy

Key Performance Metrics:

84% Accuracy in detecting Critical View of Safety after training on 2,000 surgical videos

93% Surgeon agreement with AI-generated surgical phase identification

40% Reduction in adverse events when AI quality assessment is implemented



Source: Theator Inc. "How AI Can Improve Surgical Care" (2023), Journal of Surgical Innovation (2024)

AI for Surgical Training

AI-powered video analysis is revolutionizing surgical education:

Objective Assessment

AI provides standardized evaluation of surgical technique and performance metrics

Learning Curve Acceleration

Studies show 40% faster skill acquisition with AI-guided feedback compared to traditional methods

Deliberate Practice

AI identifies specific areas for improvement, enabling targeted skill development

Expert Emulation

AI compares trainee movements and decisions with those of experienced surgeons

Intuitive Surgical clip

SimNow2 clip

The Rise of Ambient Scribes

Ambient AI scribes are transforming clinical documentation:

Passive Listening

AI systems that automatically capture and transcribe doctor-patient conversations

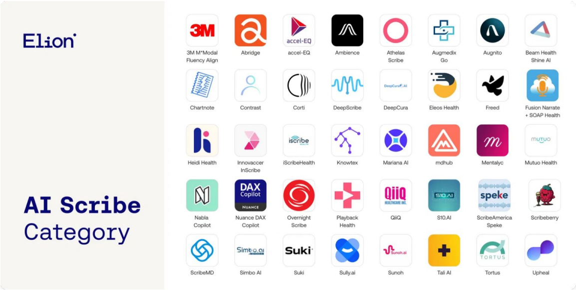
Structured Documentation

Converts natural conversation into organized clinical notes integrated with EHR systems

Time Savings

Physicians using ambient scribes report **2-3 hours** saved daily on documentation

By 2025, over 40% of healthcare organizations have implemented or are piloting ambient scribe technology to reduce clinician burnout and improve patient interaction.



Leading Ambient Scribe Solutions

Abridge

Real-time transcription with "Linked Evidence" technology

Valued at $5.3 billion (June 2025), used in 7,000+ healthcare facilities

Nuance DAX (Microsoft)

Dragon Ambient eXperience with deep EHR integration

Deployed in 150+ health systems with Epic integration

Ambience Healthcare

AI platform for documentation, coding, and clinical workflow

Selected by Cleveland Clinic after extensive evaluation process

Doximity & OpenEvidence

Free AI scribe (Doximity) and medical information platform

OpenEvidence used by 40% of US physicians, valued at $3.5B

AI for Administrative Tasks

AI is streamlining administrative workflows in healthcare:

Scheduling & Registration

Intelligent systems that optimize appointment scheduling and automate patient registration

Billing & Coding

AI-powered coding assistance that improves accuracy and reduces claim denials

Patient Communication

Automated appointment reminders, follow-ups, and medication adherence checks

Data Exchange

Streamlined requests for patient records between healthcare providers

AI in Revenue Cycle Management

AI is transforming healthcare revenue cycle management:

Claims Processing

AI predicts and prevents claim denials, reducing rejection rates by up to 30%

Clinical Documentation

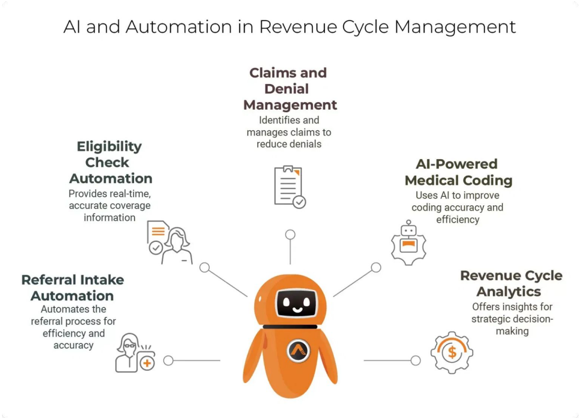
AI-powered coding assistance ensures accurate and compliant medical coding

Prior Authorization

Automated workflows reduce authorization processing time from days to hours

Predictive Analytics

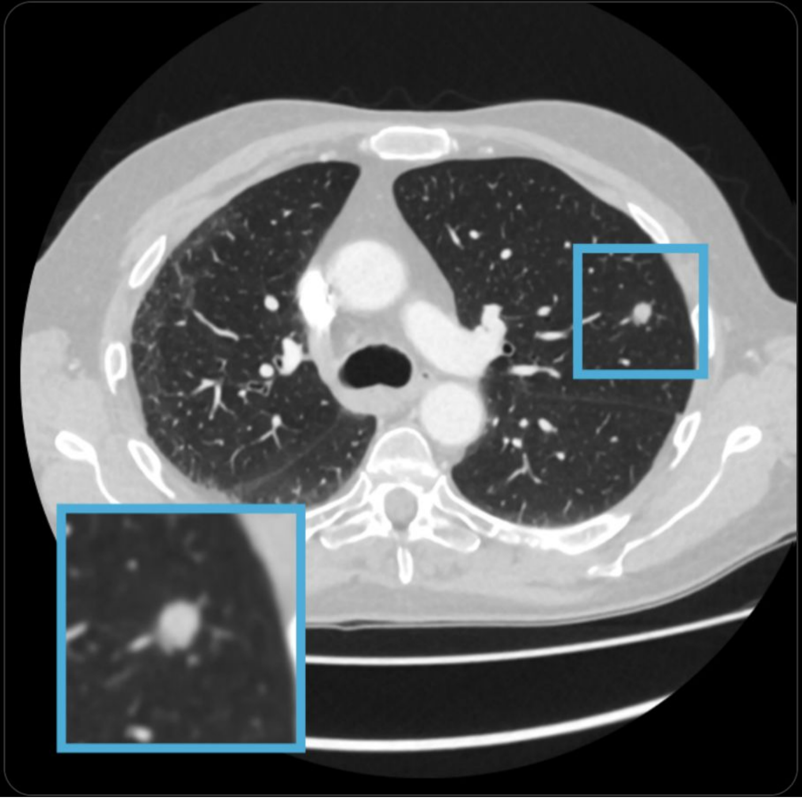
AI forecasts payment timelines and identifies optimization opportunities



Medicine is not black and white

Adam Rodman @AdamRodmanMD · Feb 8

Imagine that you are admitted to the hospital for gallstone pancreatitis. A CT scan picks up a suspicious lung nodule that could be cancer. Should you biopsy it immediately? Wait a few weeks? Perform serial imaging to make sure it doesn't grow? Or do nothing at all?



AI alone vs (AI + Doctors) vs Doctors alone

nature medicine

Article

https://doi.org/10.1038/s41591-024-03456-y

GPT-4 assistance for improvement of physician performance on patient care tasks: a randomized controlled trial

Received: 5 August 2024

Accepted: 10 December 2024

Published online: 05 February 2025

Check for updates

A list of authors and their affiliations appears at the end of the paper

While large language models (LLMs) have shown promise in diagnostic reasoning, their impact on management reasoning, which involves balancing treatment decisions and testing strategies while managing risk, is unknown. This prospective, randomized, controlled trial assessed whether LLM assistance improves physician performance on open-ended management reasoning tasks compared to conventional resources. From November 2023 to April 2024, 92 practicing physicians were randomized to use either GPT-4 plus conventional resources or conventional resources alone to answer five expert-developed clinical vignettes in a simulated setting. All cases were based on real, de-identified patient encounters, with information revealed sequentially to mirror the nature of clinical environments. The primary outcome was the difference in total score between groups on expert-developed scoring rubrics. Secondary outcomes included domain-specific scores and time spent per case. Physicians using the LLM scored significantly higher compared to those using conventional resources (mean difference = 6.5%, 95% confidence interval (CI) = 2.7 to 10.2, $P < 0.001$). LLM users spent more time per case (mean difference = 119.3 s, 95% CI = 17.4 to 221.2, $P = 0.02$). There was no significant difference between LLM-augmented physicians and LLM alone (−0.9%, 95% CI = −9.0 to 7.2, $P = 0.8$). LLM assistance can improve physician management reasoning in complex clinical vignettes compared to conventional resources and should be validated in real clinical practice. ClinicalTrials.gov registration: NCT06208423.

Large language models (LLMs) show considerable abilities in diagnostic reasoning, outperforming previous artificial intelligence (AI) models and human physicians in their ability to construct helpful differential diagnoses, explain reasoning and collect historical information from standardized patients[1–5]. LLMs have not yet been shown to perform similarly in management reasoning, which encompasses decision-making around treatment, testing, patient preferences, social determinants of health and cost-conscious care, all while managing risk[6–8].

While there is overlap, clinical reasoning is often considered to include both diagnostic and management reasoning. The study of diagnostic reasoning has a century-long history with many metacognitive frameworks and assessment methods, while management reasoning processes are a comparatively recent area of study[9–11]. Current frameworks in management reasoning include context-dependent concepts, such as shared decision-making, dynamic relationships and competing priorities between medical systems and individuals, the physician–patient relationship and time constraints inherent in modern clinical encounters[8,12–14]. Unlike diagnostic reasoning, which can be thought of as a classification task with often a single right answer, management reasoning may have no right answers and involves weighing trade-offs between inherently risky courses of action; even inaction through 'watchful waiting' is a deliberate choice with potential risks

The most radical model might be complete separation: having A.I. handle certain routine cases independently (like normal chest X-rays or low-risk mammograms), while doctors focus on more complex disorders or rare conditions with atypical features.

Early evidence suggests this approach can work well in specific contexts. A Danish study published last year found that an A.I. system could reliably identify about half of all normal chest X-rays, freeing up radiologists to devote more time to studying images that were deemed suspicious. In a landmark Swedish trial involving mammograms for more than 80,000 women, half the scans were assessed by two radiologists, as is usual. The other half were evaluated by A.I.-supported screening first, followed by additional review by one radiologist (and in rarer instances where the A.I. determined an elevated risk, by two radiologists). The A.I.-assisted approach led to the identification of 20 percent more breast cancers while reducing the overall radiologist workload almost in half.

Radiology

REVIEWS AND COMMENTARY • EDITORIAL

Beyond Assistance: The Case for Role Separation in AI-Human Radiology Workflows

Pranav Rajpurkar, PhD[1] • Eric J. Topol, MD[2]

In radiology, the prevailing view has long favored assistive artificial intelligence (AI) approaches, where human radiologists work "with" AI systems, under the belief that merging machine precision with human judgment naturally leads to improved diagnostic outcomes. However, recent evidence challenges this assumption by demonstrating that integrating AI into radiologists' workflows does not always yield the anticipated enhancements. This emerging perspective suggests that a clearly defined division of labor—where AI and radiologists assume separate responsibilities in the diagnostic process—might provide more reliable results.

In the assistive AI model, AI systems provide their analysis first, followed by radiologist review and independent interpretation. The AI outputs—typically highlighting potential findings or providing quantitative measurements—are made available to radiologists as they conduct their own evaluation. This approach assumes that having access to AI insights will help radiologists catch potential oversights and make more informed decisions, while still maintaining their autonomy in the final interpretation.

However, practical implementations of the assistive model have revealed notable challenges. Agarwal et al (1) demonstrated that when discrepancies arose between AI assessment and radiologist interpretation of chest radiographs, radiologists tended to discount the AI output—even when the algorithm was correct. This *automation neglect* is only part of the story, as radiologists may have valid reasons for disregarding AI suggestions, particularly when AI systems occasionally produce clinically implausible results (2).

Conversely, radiologists can also exhibit *automation bias*—the tendency to rely overmuch on AI suggestions even when they are incorrect. In mammography interpretation, Dratsch et al (3) found that even experienced radiologists saw their diagnostic accuracy drop substantially (from 82.3% to 45.5%) when given incorrect AI predictions. This highlights that the challenges of human-AI collaboration extend beyond automation neglect to include potentially dangerous overreliance on AI systems.

These studies suggest that the integration of AI into clinical workflows faces substantial cognitive and practical challenges that must be addressed for effective implementation.

The Role Separation Framework

At its core, role separation leverages the distinct strengths of AI and radiologists by assigning them complementary yet separate responsibilities in the diagnostic workflow. Rather than merging efforts into a single, integrated process—which can lead to issues like automation neglect and automation bias—this approach divides the workflow into segments best suited to each party's unique capabilities.

We propose three distinct models of role separation. In the *AI-first sequential model*, AI undertakes an initial portion of the workflow, handling parts of the process that can be effectively automated. Once the AI completes its portion, the radiologist takes over for expert interpretation. In the *doctor-first sequential model*, the radiologist initiates the diagnostic process using their clinical expertise, while the AI performs a distinct, complementary task that enhances the overall workflow. In the *case allocation model*, the responsibility for handling a case is determined by its characteristics. Depending on the complexity and clarity of a given case, it may be managed entirely by AI, entirely by the radiologist, or by a combination of both.

In practice, the boundaries between these approaches often blur, with many implementations involving iterative interactions rather than strict sequential processes. The implementation of AI tools in radiology workflows may need to accommodate different radiologist preferences, different institutional practices, and the specific requirements of different imaging modalities and clinical contexts.

The AI-First Sequential Model

One compelling application of the AI-first sequential model involves using AI to analyze electronic health records to prepare relevant clinical context before radiologist interpretation. In this workflow, AI systems review prior reports, laboratory results, and clinical notes to compile a focused summary of the patient's medical history and current clinical questions relevant to the imaging study.

Bhayana et al (4) evaluated the ability of GPT-4 (OpenAI) to automatically generate clinical histories for CT requisitions from clinical notes at a cancer center. The results were encouraging: Histories generated by the large language model included more critical parameters than physician-provided requisition histories, such as primary oncologic diagnosis (99.5% vs 89%), acute symptoms (15% vs 4%), and relevant surgical history (61% vs 12%). Radiologists generally preferred the AI-generated histories for image interpretation (89% vs 5%).

However, challenges remain in implementing this approach. While AI tools can gather and structure information, maintaining context across distributed information sources remains challenging. As Johri et al (5) demonstrated, the diagnostic accuracy of large language models drops substantially when reasoning across fragmented conversation turns versus reading a consolidated patient history. This limitation applies to AI systems attempting to extract and organize relevant clinical information from fragmented electronic health record documentation, suggesting that substantial improvements in context management capabilities will be necessary before these systems can be widely deployed in clinical radiology workflows.

Author affiliations, funding, and conflicts of interest are listed at the end of this article.

Radiology 2025; 316(1):e250477 • https://doi.org/10.1148/radiol.250477 • Content code: AI •

AI for Patient Engagement

AI is empowering patients to become more active participants in their healthcare:

Health Literacy

AI tools translate complex medical information into accessible, personalized explanations

Treatment Adherence

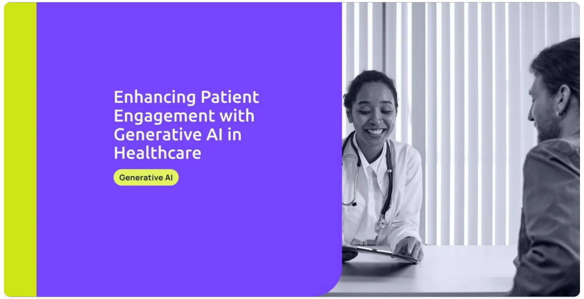
Smart reminders and personalized education improve medication compliance by up to 40%

Pre-visit Preparation

AI assistants help patients formulate questions and organize concerns before appointments

Ongoing Monitoring

AI-powered apps track symptoms and provide actionable insights between clinical visits



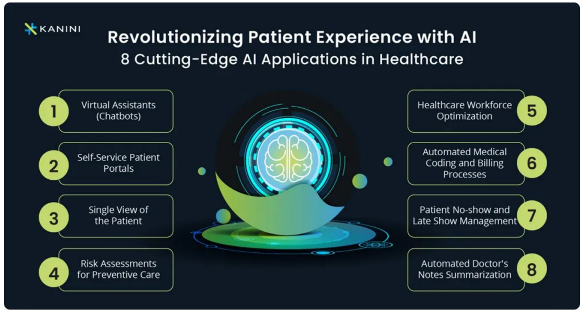
Hippocratic AI voice agent clip

Leveling Information Asymmetry

AI is transforming the traditional knowledge gap between patients and providers:

> *"When patients feel unheard, arming themselves with knowledge becomes a strategy to be taken seriously."*
>
> *— JAMA, "When Patients Arrive With Answers," 2025*

- AI tools provide patients with medical information previously accessible only to clinicians
- Shifting from "doctor knows best" to shared decision-making model
- Patients arrive better prepared for meaningful dialogue about their care



The Atlantic notes: "At its best, medicine will be a process of shared decision making, and doctors need to be prepared." This requires both common knowledge and clear communication frameworks between patients and providers.

Sources: JAMA, "When Patients Arrive With Answers," 2025; The Atlantic, "When Patients Do Their Own Research," 2024

A PIECE OF MY MIND

AI IN MEDICINE

When Patients Arrive With Answers

Kumara Raja Sundar, MD
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She came in for dizziness and described her symptoms with striking precision: "It's not vertigo, more of a presyncope kind of feeling." Then she added, "I think a tilt table test might clarify what's going on." Occasionally, a patient's questions reveal a subtle fluency in medical jargon, suggesting they might have clinical training or at least have done a lot of reading. Following the breadcrumbs, I asked gently, "Do you work in health care?" Usually, if a yes, the response is hesitant, perhaps concerned that their background might influence care. But this time, she looked down and quietly said that she had asked ChatGPT, and it mentioned tilt table testing.

Patients arriving with researched information is not new. They have long brought newspaper clippings, internet search results, or notes from conversations with family. Potential solutions passed along in WhatsApp threads have at times been an integral part of my clinical conversations. Information seeking *outside* the health care setting has always been part of the landscape of care.

But something about this moment feels different. Generative artificial intelligence (AI), with tools like ChatGPT, offers information in ways that feel uniquely conversational and tailored. Their tone invites dialogue. Their confidence implies competence. Increasingly, patients are bringing AI-generated insights into my clinic and are sometimes confident enough to challenge my assessment and plan.[1]

I heard these tools were helpful, but I understood their appeal only after using them myself. Recent studies have bolstered this claim: large language models (LLMs) show surprising strength in reasoning and relational tone.[2] After seeing it firsthand, my reaction was simple: "Man, I get why my patients like it."

Yet comparing LLMs to clinicians feels inherently unfair. Clinicians often work under pressure with rushed visits and overflowing inboxes as health care systems demand productivity and performance.[3] My concern is that research is comparing clinicians who are not given the luxury of performing their best under strained systems with the inexhaustible resources of generative AI. That is not a fair fight, but it is reality. I find patients may seek clear answers, but even more, they want to feel recognized, reassured, and truly heard. Unfortunately, under the weight of competing demands, that is what often slips for me—not just because of systemic constraints but also because I am merely human.

Despite generative AI's capabilities, patients still come to see me. The tools they use usually offer confident suggestions but eventually defer with some version of a familiar disclaimer: "Consult a medical professional." Accountability for care still rests with the clinician. Both in terms of liability, who will take the blame if something goes wrong or someone gets hurt, and in terms of responsibility, who will place the orders for diagnostic plans and prescriptions for care. Clinicians remain the gatekeepers. In practice, this means navigating patient requests like a tilt-table test for intermittent dizziness—tests that are not unusual but may not be appropriate at a specific stage of care. I find myself explaining concepts like overdiagnosis, false-positives, or other risks of unnecessary testing. At best, the patient understands the ideas, which may not resonate when one is the person experiencing symptoms. At worst, I sound dismissive. There is no function that tells ChatGPT that clinicians lack routine access to tilt-table testing or that echocardiogram appointments are delayed due to staffing shortages. I have to carry those constraints into the examination room while still trying to preserve trust.

> Seeing AI-informed visits as opportunities for deeper dialogue rather than threats to clinical authority may sound aspirational, but it reflects a necessary shift.

When I speak with medical students, I notice a different kind of paternalism creeping in. And I have caught it in my inner monologue even if I do not say it aloud. The old line, "They probably WebMD'd it and think they have cancer," has morphed into the newer, just-as-dismissive line, "They probably ChatGPT'd it and are going to tell us what to order." It often reflects defensiveness from clinicians rather than genuine engagement and carries an implicit message: We still know best. It is an attitude that risks eroding the sacred and fragile trust between clinicians and patients. It reinforces the feeling that we are not "in it" with our patients and are truly gatekeeping rather than partnering. Ironically, that is often why I hear patients turn to LLMs in the first place.

One patient said plainly, "This is how I can advocate for myself better." The word *advocate* struck me. It is what one does when trying to convince someone with more power. I know clinicians still hold the keys to ordering tests, referrals, and treatment plans, but there is a sense of preparing for a fight that the word *advocate* conveyed. When patients feel unheard, arming themselves with knowledge becomes a strategy to be taken seriously. I vividly recall another patient who, following a pregnancy loss, arrived with a specific set of evaluations she wanted. Her request was layered with grief, prior dismissal, and the quiet hope that being prepared might finally lead to being heard. She did not need the

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Section Editor: Preeti Malani, MD, MSJ, Deputy Editor.

AI can (rarely) make mistakes

Gemini

2.5 Pro

Invite a friend

PRO

2. Side Effect Profile and Patient's Medical History

This is the most critical factor for personalizing the choice. AIs and Tamoxifen have very different potential side effects, so the "better" drug for an individual depends on their personal health history.

- **Aromatase Inhibitors (AIs) are generally preferred for patients who do NOT have:**
  - **Significant bone health issues:** The main side effect of AIs is an increased risk of **bone thinning (osteoporosis) and fractures**. They can also cause **musculoskeletal symptoms** like joint pain and stiffness. Therefore, a patient with severe, pre-existing osteoporosis might be a better candidate for Tamoxifen.
- **Tamoxifen is generally preferred for patients who have:**
  - **A history of or high risk for blood clots:** Tamoxifen carries a small but serious increased risk of **deep vein thrombosis (blood clots)**.
  - **Uterine issues:** Tamoxifen also slightly increases the risk of **uterine cancer**.
  - **Severe osteoporosis:** Since Tamoxifen does not cause bone loss (and can even slightly strengthen bones in postmenopausal women), it is a better option for those with significant bone health concerns.

Gemini

2.5 Pro

This does not make any sense. Explain this

"Tamoxifen is generally preferred for patients who have:

A history of or high risk for blood clots: Tamoxifen carries a small but serious increased risk of deep vein thrombosis (blood clots)"

Show thinking

You are absolutely right to point that out. My previous response contained a serious error and was very confusing. I sincerely apologize for the mistake and the lack of clarity. Thank you for catching it.

Let me provide the correct information. The statement should have been the exact opposite.

The choice between an Aromatase Inhibitor (AI) and Tamoxifen often depends on which drug's side effect profile is safer for the individual patient.

Here is the corrected and clarified explanation:

- An **Aromatase Inhibitor (AI)** would be the preferred choice for a patient with a **history of or high risk for blood clots**. This is because AIs do not carry this risk.
- **Tamoxifen** would be **avoided** in that same patient. A personal history of blood clots is a major contraindication (a reason not to use a drug) for Tamoxifen because it carries a small but serious increased risk of deep vein thrombosis (blood clots) and pulmonary embolism.

To summarize the decision correctly:

- If a patient has a **history of blood clots**, an **AI is preferred**.
- If a patient has **severe osteoporosis**, **Tamoxifen may be preferred** (because AIs can worsen bone loss).

Is AI making doctors better ?

(can it, unintentionally, make them worse ?)

THE LANCET
Gastroenterology & Hepatology

ARTICLES · Online first, August 12, 2025

Endoscopist deskilling risk after exposure to artificial intelligence in colonoscopy: a multicentre, observational study

Krzysztof Budzyń, MD [a,b] · Marcin Romańczyk, MD [a,b] · Diana Kitala, PhD [c] · Paweł Kołodziej, MD [d] · Marek Bugajski, MD [e] · Hans O Adami, MD [f,g] · et al. Show more

Summary

Background

It is not known if continuous exposure to artificial intelligence (AI) changes endoscopists' behaviour when conducting colonoscopy. We assessed how endoscopists who regularly used AI performed colonoscopy when AI was not in use.

Methods

We conducted a retrospective, observational study at four endoscopy centres in Poland taking part in the ACCEPT (Artificial Intelligence in Colonoscopy for Cancer Prevention) trial. These centres introduced AI tools for polyp detection at the end of 2021, after which colonoscopies had been randomly assigned to be conducted with or without AI assistance according to the date of examination. We evaluated the quality of colonoscopy by comparing two different phases: 3 months before and 3 months after AI implementation. We included all diagnostic colonoscopies, excluding those involving intensive anticoagulant use, pregnancy, or a history of colorectal resection or inflammatory bowel disease. The primary outcome was change in adenoma detection rate (ADR) of standard, non-AI assisted colonoscopy before and after AI exposure. Multivariable logistic regression was done to identify independent factors affecting ADR.

Findings

Between Sept 8, 2021, and March 9, 2022, 1443 patients underwent non-AI assisted colonoscopy before (n=795) and after (n=648) the introduction of AI (median age 61 years [IQR 45–70], 847 [58·7%] female, 596 [41·3%] male). The ADR of standard colonoscopy decreased significantly from 28·4% (226 of 795) before to 22·4% (145 of 648) after exposure to AI, corresponding with an absolute difference of –6·0% (95% CI –10·5 to –1·6; p=0·0089). In multivariable logistic regression analysis, exposure to AI (odds ratio 0·69 [95% CI 0·53–0·89]), male versus female patient sex (1·78 [1·38–2·30]), and patient age ≥60 years versus <60 years (3·60 [2·74–4·72]) were the independent factors significantly associated with ADR.

Interpretation

Continuous exposure to AI might reduce the ADR of standard non-AI assisted colonoscopy, suggesting a negative effect on endoscopist behaviour.

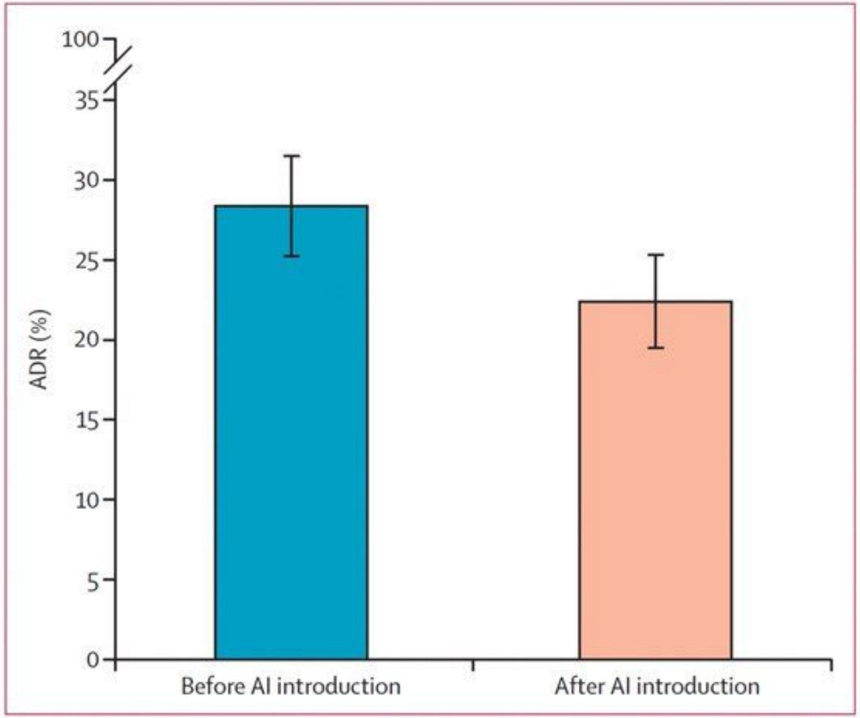


Figure 1: **Change in ADR with standard, non-AI assisted colonoscopy before and after introduction of AI for polyp detection**

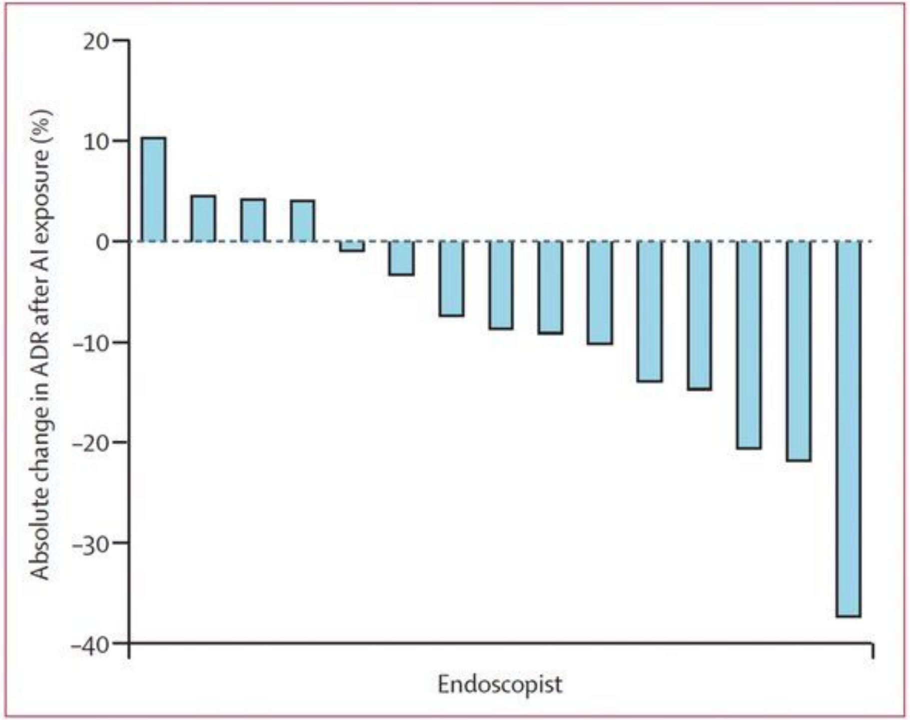


Figure 2: **Endoscopist-level change in ADR with standard, non-AI assisted colonoscopy after introduction of AI in colonoscopy at each centre**

AI Agents in Oncology: Projects from Our Lab

AI-Powered Virtual Molecular Tumor Board

Less than 5% of India’s 1.4M annual cancer patients receive expert multidisciplinary review

Access Gap: 4x fewer oncologists per capita than the US; tier-2/3 cities worst affected

Staging Ambiguity: Uncertain staging can mean the difference between curable and terminal classification

Time Pressure: Diagnosis-to-treatment timelines stretch from days to weeks without MDT access

AI-powered virtual MTB reduces diagnosis-to-recommendation time from weeks to minutes

Virtual MTB: Architecture & Capabilities

Bringing expert-level oncology recommendations to underserved regions

Evidence-Based Recommendations: Aligned with NCCN, ESMO international treatment standards

Local Contextualization: Accounts for regional drug availability, costs, and insurance coverage in India

Scalable Deployment: Targets tier-2/3 cities and rural areas where specialist access is limited

Clinical Validation: Evaluated on synthetic oncology cases; seeking clinical validation partners

Onco-Seg: AI-Powered Tumor & Organ Segmentation

Radiation oncologists spend 2–4 hours per patient manually contouring 30+ organs-at-risk

One Model, Many Tasks: Single unified model across CT, MRI, ultrasound, dermoscopy, endoscopy, PET-CT, X-ray, histology

Sub-Second Inference: GPU-based segmentation enabling real-time clinical workflows

Parameter Efficient: LoRA fine-tuning updates <5% of SAM3's 848M parameters

Clinical Deployment Ready: Interactive sidecar for diagnostic radiology + silent assistant for automated contouring

Trained on 98,000+ cases across 35 datasets and 8 imaging modalities

Onco-Seg: SAM3 Adaptation Pipeline

Adapting Meta's Segment Anything Model 3 for clinical oncology workflows

Foundation Model: Meta SAM3 (848M params) with LoRA adapters (rank=16, alpha=32)

Training Strategy: Sequential checkpoint chaining across 8 phases; combined Dice + Focal loss

Interaction Modes: Point prompts, bounding boxes, text prompts, 3D slice propagation

Deployment: napari plugin with DICOM-RT Structure Set export for clinical integration

Dice >0.65 across liver CT, breast ultrasound, polyp detection; 80–90% time savings vs manual contouring

Onco-Shikshak: AI-Augmented Oncology Education

NCCN alone maintains 76+ guideline documents updated multiple times annually — how do learners keep up?

Virtual Tumor Board: 6 specialist AI agents (pathology, radiology, surgical/medical/radiation oncology, supportive care) deliberate on cases

Morning Report: Socratic dialogue with expertise-adaptive difficulty via Item Response Theory

Spaced Repetition: FSRS v4 scheduling with confidence calibration across cancer domains

Clinic Day Simulation: Multi-patient management under time pressure with NCCN-grounded feedback

7 guideline corpora + 2 textbooks | 6 AI agents | 18 clinical cases

Onco-Shikshak: RAG + Learning Science

Every AI response grounded in authoritative clinical sources with automated citation extraction

RAG Foundation: Gemini File Search over NCCN, ESMO, ASTRO, ACR, CAP, ClinVar/CIViC guideline corpora

Automated Citation: Grounding metadata ensures every recommendation traces back to source guidelines

Expertise Adaptation: IRT-based ability estimation adjusts content difficulty in real-time

Learning Science: Maps to spacing effect, retrieval practice, expertise reversal, desirable cognitive friction

Onco-TTT: AI-Powered Cancer Hypothesis Generation

Bridging the gap between clinical observations and research hypotheses in oncology

Entity Extraction: GLiNER2-based recognition of 10 biological entity types (genes, diseases, drugs, pathways, mutations, etc.)

Knowledge Graph: Automated construction integrating NER with OpenTargets GraphQL enrichment

Multi-Source Validation: Evidence dashboard cross-referencing DepMap, cBioPortal, GTEx, ClinicalTrials.gov

Literature Integration: Semantic Scholar search + CELLxGENE single-cell atlas projection

Onco-TTT: From Query to Hypothesis

Natural language query → Entity extraction → Knowledge graph → OpenTargets enrichment → Hypothesis generation → Evidence dashboard

Protein Structure: AlphaFold integration for structural analysis of target proteins

Patent Landscape: USPTO PatentsView queries for intellectual property context

Cell Line Selection: Cellosaurus-powered recommendations for experimental validation

Protocol Generation: CRISPR guide suggestions for functional validation experiments

Palli Sahayak: Voice AI for Palliative Care

Only 1–2% of 10M+ patients needing palliative care in India have access

Voice-First Design: Toll-free phone calls and WhatsApp bot; no literacy requirement

15+ Indian Languages: Hindi, Bengali, Tamil, Marathi, Punjabi, Malayalam, Telugu, Kannada, Gujarati, English

Clinically Validated: Built on palliative care case vignettes from Pallium India and Max Healthcare

Privacy-First: No PHI stored; sessions expire after 24 hours; HIPAA-aligned practices

Targets 10M+ patients and 1M+ ASHA workers across India

Palli Sahayak: Architecture & Impact

Grand Challenges India Award (Nov 2024) — Bill & Melinda Gates Foundation / BIRAC-DBT

Hybrid RAG Pipeline: Vector storage + Neo4j knowledge graph for clinical knowledge retrieval

Safety Layer: Dosage validation, hallucination detection, source citation, emergency referral

Voice Stack: Bolna AI + Twilio STT + Gemini Live for natural multilingual conversations

Digital Public Good: MIT License, UN SDG aligned (Goals 3 & 10)

Future Directions and Challenges

As AI continues to transform healthcare, several key challenges must be addressed:

Privacy and Security

Ensuring patient data protection while enabling AI systems to learn from clinical data

Ethical Considerations

Addressing bias, transparency, and accountability in AI healthcare applications

Clinical Validation

Establishing rigorous standards for evaluating AI performance in real-world settings

Workforce Adaptation

Training healthcare professionals to effectively collaborate with AI systems

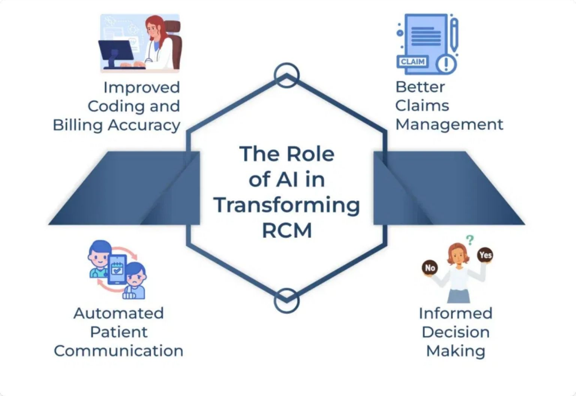
Conclusion

AI agents are transforming healthcare across multiple domains:

- **Medical Education:** Enhancing training through virtual patients, empathy development, and diagnostic skill building
- **Clinical Practice:** Supporting radiologists, improving surgical quality, and reducing documentation burden
- **Healthcare Operations:** Streamlining administrative tasks and optimizing revenue cycle management
- **Patient Experience:** Empowering patients with information and improving engagement in care decisions

Oncology Applications:
AI-powered tumor boards, segmentation, education, hypothesis generation, and palliative care voice assistants

> *"AI in healthcare is not about replacing human expertise, but augmenting it—creating a future where technology enhances the human elements of care."*



Thank you for your attention
August 2025

NEJM AI 2025;2(8)
DOI: 10.1056/AIp2500260

PERSPECTIVE

The Student Physician–Patient–AI Relationship

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Received: March 13, 2025; Revised: May 17, 2025; Accepted: June 2, 2025; Published: July 16, 2025

Abstract

As first-year medical students participating in an ambient AI scribe pilot program for our first-ever standardized patient interviews, we were struck by the challenge of interviewing patients, but even more so by the AI scribe's remarkably comprehensive output. We grappled with feelings of inadequacy and admiration, leading to unexpected competition between us and the AI scribe. Throughout many simulated patient interviews, we attempted to outdo the AI that had stunned us, making substantial growth but still falling short. A real-world patient encounter would eventually uncover our unique capabilities. In the competitive relationship that we developed with AI, we witnessed the potential of improving our skill sets in union with AI and learned how to leverage our competitive advantage. (Funded by National Institutes of Health Medical Scientist Training Program; training grant number, T32GM136651.)

Solemnly gazing at his feet, Franklin explained what had brought him to seek care: a pervasive loss of interest in his most meaningful hobbies, including reading, gardening, and spending time with friends. Tears welled in his eyes as he explained how this had created emotional distance between him and his wife, Evelyn, who had implored him to go see the doctor. Despite this being a classic presentation for major depressive disorder, as first-year medical students, we struggled to make sense of Franklin's story, nervously fumbling over our words and scribbling notes. Like the countless medical students who came before us, we were participating in our very first standardized patient interviews: a lower-stakes encounter to practice communication skills before caring for real patients in the hospital. However, our experience was fundamentally different.

Before Franklin entered, our session facilitator pulled out an innocuous smartphone and loaded an application. Instead of only Franklin and us being in the room, an ambient AI scribe technology would quietly listen beside us; one touted to improve the physician–patient relationship through simplifying clerical work. It would follow along throughout our conversations with Franklin and then independently write a full visit summary and medical note. Our facilitator's passionate description of this technology made it clear this tool was distinct from others in our widening medical student tool kits. Our first encounter had an AI scribe sitting by our side.

Disinterest, emotional distance, loss of appetite — we jotted down Franklin's concerns as his story unfolded. All the while, our eyes darted between him and our interview checklist, struggling to hold his gaze while scrambling to identify our next question. Gradually, our notes began to take shape, forming a fuller account of his story — even noting his declining

Mr. Lee and Mr. Link contributed equally to this article, and Drs. Hersh and Angoff contributed equally to this article.

Note: Identifying details regarding the patient encounter have been changed to protect patient privacy.

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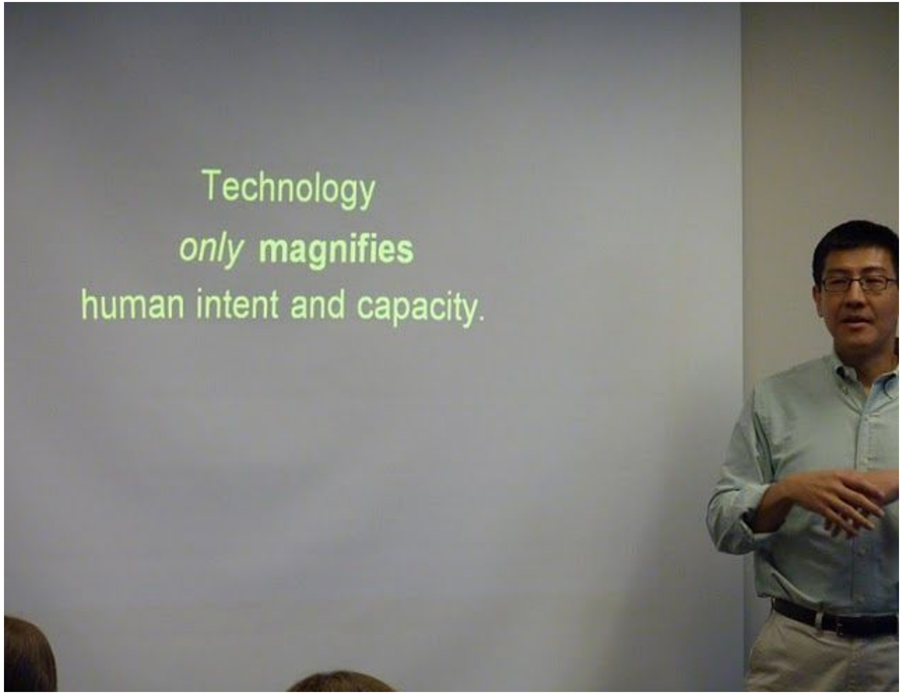
Technology & ai is not a panacea !



Where to learn more ?

http://mlformds.com/

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